

A Polynomial-Time Noisy Learner with Square-Root Regret

Research note, 4 September 2026

This is a separate derivation for the setting of `efficient_imprecise_bandits.typ`.

The fixed-block $T^{\frac{2}{3}}$ bound can be improved to $\tilde{O}(\sqrt{T})$. The learner below does not need to know the true hypothesis, its residual subspace, or the non-tangency constant S . It uses a quadratic penalty on lifted arm-outcome vectors, and stops each block when a single empirical quadratic form crosses a threshold. Its planning problem has a fixed number of quadratic constraints.

The dependence on S^{-1} and reward scales is explicit below. As with the existing manuscript, a bound whose prefactor is polynomial in description length requires the relevant numerical parameters to be polynomially bounded; finite binary descriptions alone do not bound their magnitudes by a polynomial.

1. Normalization and statement

Translate and rescale both balls so that $X = \{x : \|x\|_2 \leq 1\}$ and $D = \{y : \|y\|_2 \leq 1\}$. Continue to write the transformed reward as $r(x, y) = a^T x + b^T y + c$. Thus the transformed $\|b\|$ is the original $R_D \|b\|$. The non-tangency constant is unchanged by this translation and uniform rescaling of outcome coordinates. Zero-radius balls give trivial special cases.

Let

$$q := 1 + d_X + d_D, \quad w(x, y) := (1, x, y) \in \mathbb{R}^q, \quad W := \sqrt{3}.$$

For $\delta \in (0, 1)$, define quantities used only in the analysis:

$$J_T := \left\lceil q \log_2 \left(1 + \frac{3T}{q} \right) \right\rceil,$$

$$h_T^2 := 8d_D \log \left(\frac{2d_D T^2}{\delta} \right),$$

$$B_T^2 := 1 + J_T h_T^2,$$

$$G := \sqrt{3}(1 + S^{-1})\|b\|.$$

Claim. With $\lambda = \lceil \sqrt{T} \rceil$, the learner below has, with probability at least $1 - \delta$, realized regret at most

$$TV^* - \sum_{t=1}^T r(x_t, y_t) \leq \lambda (1 + \sqrt{3})^2 (J_T + 1) + \frac{G^2 B_T^2 T}{4\lambda} + T\varepsilon,$$

when every planning call has additive error at most ε . In particular, $\delta = (T + 1)^{-2}$ and $\varepsilon = (T + 1)^{-2}$ give

$$R_T = \tilde{O} \left(\left[q + (1 + S^{-1})^2 \|b\|^2 q d_D \right] \sqrt{T} \right),$$

with a vanishing additional contribution from the failure event. The policy runs in time polynomial in the input length and T , under the same weak optimization model used in the manuscript. Neither S nor B_T is an input to the policy.

2. Learner

Initialize $V = I_q$. At the start of each block:

1. Define

$$U_V(x) := \min_{y \in D} [r(x, y) + \lambda w(x, y)^T V^{-1} w(x, y)].$$

Choose $x \in X$ with $U_V(x) \geq \max_{x' \in X} U_V(x') - \varepsilon$.

2. Keep playing this arm. After n observations in the block, set

$$\bar{y}_n := \frac{1}{n} \sum_{i=1}^n y_i, \quad \bar{w}_n := (1, x, \bar{y}_n).$$

End the block at the first n satisfying

$$n \bar{w}_n^T V^{-1} \bar{w}_n \geq 1.$$

3. At such an endpoint, update

$$V \leftarrow V + n \bar{w}_n \bar{w}_n^T$$

and begin a new block. Stop at the horizon, even if the final block has not crossed the threshold.

There is no anchor-sampling phase. The adversary may change its distribution after every observation, including within a block.

3. Proof

3.1. One fixed subspace contains every lifted conditional mean

Let $N = \ker C_*$, and let Q be the orthogonal projector onto $N^\perp$. The minimum-norm solution of the true affine mean constraints is an affine map $f(x) = Ax + d \in N^\perp$. It exists for every arm because every true fiber is nonempty. Consequently

$$K^*(x) = (Ax + d + N) \cap D.$$

Since the minimum-norm solution has norm at most that of any feasible mean, $\|Ax + d\| \leq 1$ on X . Evaluating at 0 and at opposite unit vectors gives $\|d\| \leq 1$ and $\|A\|_{\text{op}} \leq 1$.

Define the linear map and its kernel

$$F(s, x, y) := Qy - Ax - sd, \quad L := \ker F.$$

Every conditional mean m_t satisfies $w(x_t, m_t) \in L$, and $\|F\|_{\text{op}} \leq \sqrt{3}$. Hence, for $x \in X$ and $y \in D$,

$$\text{dist}(y, Ax + d + N) = \|F(1, x, y)\| \leq \sqrt{3} \text{dist}(w(x, y), L).$$

The manuscript's non-tangency inequality then gives

$$\text{dist}(y, K^*(x)) \leq \sqrt{3}(1 + S^{-1}) \text{dist}(w(x, y), L). \quad (1)$$

All these objects are for analysis only.

3.2. There are logarithmically many completed blocks

For a completed block, the determinant update is

$$\det(V + n \bar{w} \bar{w}^T) = \det(V)(1 + n \bar{w}^T V^{-1} \bar{w}) \geq 2 \det(V).$$

After J completed blocks, $\det V \geq 2^J$. Every block average satisfies $\|\bar{w}\| \leq \sqrt{3}$, so $\text{tr } V \leq q + 3T$. The arithmetic-geometric mean inequality gives

$$2^J \leq \det V \leq \left(1 + 3 \frac{T}{q}\right)^q,$$

and therefore $J \leq J_T$.

The stopping rule also controls overshoot. Within a block put $S_n := \sum_{i=1}^n w(x, y_i)$ and use the norm induced by its fixed V^{-1} . At the first crossing, for $n \geq 2$,

$$\begin{aligned} \frac{\|S_n\|_{V^{-1}}}{\sqrt{n}} &\leq \sqrt{\frac{n-1}{n}} \frac{\|S_{n-1}\|_{V^{-1}}}{\sqrt{n-1}} + \frac{\|w(x, y_n)\|_{V^{-1}}}{\sqrt{n}} \\ &\leq 1 + \sqrt{3}. \end{aligned}$$

Here $V \succeq I$. The same upper bound holds if $n = 1$, or for a final uncrossed block. Thus every block used in the regret calculation satisfies

$$n\bar{w}_n^T V^{-1} \bar{w}_n \leq (1 + \sqrt{3})^2. \quad (2)$$

3.3. Noise has little energy perpendicular to the true subspace

Write m_t for the conditional mean after the adversary has chosen the round's distribution. The coordinates of $y_t - m_t$ are martingale differences bounded in absolute value by 2. Azuma-Hoeffding and a union bound over coordinates and all at most T^2 deterministic time intervals give, with probability at least $1 - \delta$,

$$n\|\bar{y} - \bar{m}\|^2 \leq h_T^2 \quad (3)$$

simultaneously on every interval. This includes blocks with adaptive starting times and endpoints; no optional-stopping assumption about their averages is necessary.

Fix a unit vector $u \in L^\perp$. In each completed block the arm is constant, so $(1, x, \bar{m}) \in L$ and

$$n(u^T \bar{w})^2 = n(u^T(0, 0, \bar{y} - \bar{m}))^2 \leq h_T^2.$$

It follows simultaneously for all current matrices and all such u that

$$u^T V u \leq 1 + J_T h_T^2 = B_T^2.$$

By the Cauchy-Schwarz inequality in the V and V^{-1} norms,

$$\text{dist}(w, L) = \sup_{u \in L^\perp, \|u\| \leq 1} u^T w \leq B_T \sqrt{w^T V^{-1} w}. \quad (4)$$

3.4. The quadratic penalty gives sufficient optimism without knowing S

For any $x \in X$ and $y \in D$, project y to $K^*(x)$ and use Equation 1 and Equation 4. This yields

$$v^*(x) \leq r(x, y) + G B_T \sqrt{w(x, y)^T V^{-1} w(x, y)}.$$

Writing $\ell = \sqrt{w^T V^{-1} w}$ and completing the square,

$$\lambda \ell^2 - G B_T \ell \geq -\frac{G^2 B_T^2}{4\lambda}.$$

Therefore

$$U_V(x) \geq v^*(x) - \zeta_T, \quad \zeta_T := \frac{G^2 B_T^2}{4\lambda}. \quad (5)$$

Although the unknown geometric constant appears in this analysis, the quadratic penalty itself only uses $\lambda = \lceil \sqrt{T} \rceil$.

For a block played at the approximate maximizer x , Equation 5 gives $U_V(x) \geq V^* - \zeta_T - \varepsilon$. Its empirical outcome average belongs to D , so the definition of U_V also gives

$$U_V(x) \leq r(x, \bar{y}) + \lambda \bar{w}^T V^{-1} \bar{w}.$$

Multiplying by the block length and using affineness of reward and Equation 2, with $\mathcal{B}$ denoting the block,

$$\sum_{i \in \mathcal{B}} (V^* - r(x, y_i)) \leq \lambda (1 + \sqrt{3})^2 + n(\zeta_T + \varepsilon).$$

Summing over at most $J_T + 1$ blocks proves the claim. On the failure event, realized regret is at most the reward range times T ; with the stated choice of δ , its contribution to expected regret is $O\left(\frac{C_r}{T}\right)$.

4. Polynomial-time planning

The remaining task is to maximize U_V over the arm ball. The identity

$$\lambda w^T V^{-1} w = \max_z \left[z^T w - \frac{z^T V z}{4\lambda} \right]$$

has maximizer $z = 2\lambda V^{-1} w$. Since $V \succeq I$ and $\|w\| \leq \sqrt{3}$, this maximizer has norm at most $2\sqrt{3}\lambda$. We can therefore restrict z to $\|z\| \leq 4\lambda$, uniformly in x, y .

For each fixed x , the expression is affine in y and concave in z , and both domains are compact and convex. The minimax theorem allows the minimum over y and maximum over z to be exchanged. Partitioning $z = (z_0, z_X, z_D)$ gives

$$U_V(x) = a^T x + c + \max_{\|z\| \leq 4\lambda} \left[z_0 + z_X^T x - \|b + z_D\| - \frac{z^T V z}{4\lambda} \right].$$

Consequently the planning step is the following single QCQP:

$$\max_{x, z, s} \quad a^T x + c + z_0 + z_X^T x - s - \frac{z^T V z}{4\lambda}$$

subject to

$$\|x\|^2 \leq 1, \quad \|z\|^2 \leq 16\lambda^2, \quad \|b + z_D\|^2 \leq s^2, \quad 0 \leq s \leq H,$$

where $H = 1 + \|b\|_1 + 4\lambda$ is a rational upper bound. Add the redundant ellipsoidal constraint

$$\|x\|^2 + \frac{\|z\|^2}{16\lambda^2} + \frac{s^2}{H^2} \leq 3.$$

There are six quadratic inequalities, independently of the dimensions, and one is strictly convex in all variables. Thus the weak polynomial-time QCQP result already used in the manuscript applies: Bienstock, *A Note on Polynomial Solvability of the CDT Problem* (2016).

To repair a weakly feasible solution, project x and z onto their balls and set $s = \|b + z_D\|$. Constraint violations at most τ require $O(\sqrt{\tau})$ movements in these variables, or a one-sided increase of that order in s . On this bounded domain the objective is Lipschitz with a polynomially bounded constant: $\lambda \leq \sqrt{T} + 1$ and $\|V\| \leq 1 + 3T$. The repaired objective therefore loses at most $\text{poly}(\text{data}, T)\sqrt{\tau}$. Choosing inverse-polynomial internal tolerance returns a feasible arm with the required additive planning accuracy. No emptiness tests, dual multiplier conditioning assumptions, or estimates of S are needed.

5. Numerical implementation

The displayed policy uses exact arithmetic to keep the proof readable. Finite precision can be handled while preserving the same bound. Round observed outcomes to rational points inside D within error $\tau_y = (T + 1)^{-4}$, and use those points in the block averages and updates. Then Equation 3 holds with h_T^2 replaced by $2h_T^2 + 2T\tau_y^2$. The norm bound and convex membership in D remain exact, and the discrepancy in accumulated reward is at most $\|b\|T\tau_y$.

Choose rational feasible arms to the required planning accuracy on a common grid. The surrogate is Lipschitz in x with constant at most $\|a\| + 2\sqrt{3}\lambda$, so this additional rounding is inexpensive. The stored matrix is then rational and positive definite, with eigenvalues between 1 and $1 + 3T$. Its updates and threshold comparisons can be carried out exactly with polynomial bit complexity. Since there are only $O(q \log T)$ updates, sums of the rational block contributions have polynomial bit length. All numerical errors can be absorbed into the stated planning error and negligible reward perturbation.